FLEX-2 Engineering Requirements Document

Fluids and Combustion Facility (FCF) Flame Extinguishment Experiment 2 (FLEX-2)

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Date	**Signature**
02/16/10	*/s/ Lori Yost*

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National Aeronautics and Space Administration
John H. Glenn Research Center
Exploration Systems Division
Cleveland, Ohio 44135

PREFACE

The National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation mini-facility for conducting droplet combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Multi-user Droplet Combustion Apparatus (MDCA), resides in Combustion Integrated Rack (CIR), a component of the Fluids and Combustion Facility (FCF) on-board the ISS. The Flame Extinguishment Experiment 2 (FLEX-2) uses the MDCA platform to conduct droplet combustion tests.

This document will define the engineering and science requirements necessary to successfully complete the FLEX-2 experiments.

SIGNATURE PAGE

FLEX-2 ENGINEERING REQUIREMENTS DOCUMENT FOR THE FLUIDS COMBUSTION FACILITY (FCF) FLAME EXTINGUISHMENT EXPERIMENT 2 (FLEX-2)

Prepared By: */s/ Jesse Robbins* **Date:** *02/05/10*
Jesse Robbins
FLEX-2 Systems Engineering Lead
ZIN Technologies, Inc.

Approved By: */s/ M. O'Toole* **Date:** *02/05/10*
Martin O'Toole
FLEX-2 Project Lead
ZIN Technologies, Inc.

Approved By: */s/ Brian Finley* **Date:** *02/05/10*
Brian Finley
SpaceDOC Manager
ZIN Technologies, Inc.

Approved By: */s/ Michael Hicks* **Date:** *02/16/10*
Michael Hicks
FLEX-2 Project Scientist
NASA Glenn Research Center

REVISION PAGE

FLEX 2 ENGINEERING REQUIREMENTS DOCUMENT FOR THE FLUIDS COMBUSTION FACILITY (FCF) FLAME EXTINGUISHMENT EXPERIMENT 2 (FLEX-2)

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1.0 INTRODUCTION

1.1 Purpose

This document will define the engineering requirements necessary to successfully complete the science and safety requirements. This document will be used by the FLEX-2 design engineers to develop the systems necessary for FLEX-2. Interface and carrier requirements levied on FLEX-2 are documented in the CIR-MDCA Interface Control Document (FCF-ICD-CIR-MDCA). FLEX-2 will use the existing Multi-User Droplet Combustion Apparatus (MDCA) hardware and software platform already developed as a baseline for meeting many of the requirements found in this document. Safety requirements are documented in the CIR/MDCA Integrated Hazard Reports and the MDCA Hazard Reports (Re-Flight).

1.2 Scope

This FLEX-2 Engineering Requirements Document (ERD) serves to document two sets of requirements: Science Requirements and Derived Engineering Requirements. The FLEX-2 Science Requirements Document (SRD) is created by the experiment's Principle Investigators (PI's) and defines the requirements that are necessary to accomplish the FLEX-2 science objectives. The purpose of this document is to provide traceability to the SRD and to formalize these requirements. Also, it defines the engineering requirements derived to establish the capabilities for a system that achieves the science requirements in conjunction with existing CIR and MDCA capabilities.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents specified herein, the requirements of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. In the event of a conflict between this document and higher-level documents, the higher-level documents shall take precedence over this document.

All documents used, applicable or reference, are to be the versions defined in the current version of the SpaceDOC contract. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The documents in the following table are applicable to the FLEX-2 Project to the extent specified herein.

FCF-ICD-CIR-MDCA	CIR FLEX-2 Interface Control Document
FLX-PLN-0002	FLEX-2 Science Engineering Verification Plan
N/A	Science Requirements Document, Fundamental Studies in Droplet Combustion and Flame Extinguishment in Micro-gravity (FLEX-2) December 6, 2009
MDC-REQ-0005	MDCA Engineering Requirements Document
NPR 6000.1	Requirements for Packaging, Handling, and Transportation for Aeronautical and Space Systems, Equipment, and Associated Components

2.3 Reference Documents

The documents in the following table are provided only as reference material for background information and are not imposed as requirements.

FCF-IA-MV-MDCA	Flame Extinguishment Experiment Integration Agreement - Main Body

FCF-ICD-0076	FCF Software Interface Control Document
KHB 1700.7	Space Shuttle Payload Ground Safety Handbook
MSFC-HDBK-527/JSC 09604	Materials Selection List for Space Hardware Systems
MSFC-SPEC-522	Design Criteria for Stress Corrosion Cracking
MDC-DOC-1364	MDCA Baseline Systems Description
NSTS 13830	Implementation Procedure for NSTS Payloads System Safety Requirements
NSTS 14046	Payload Verification Requirements
NSTS 1700.7	Safety Policy and Requirements for Payloads Using the Space Transportation System
NSTS 1700.7 ISS Addendum	Safety Policy and Requirements for Payloads Using the International Space Station
NSTS 18798A MA2-95-048	Interpretation of NSTS Payload Safety Requirements
SSP 50005	International Space Station Flight Crew Integration Standard
SSP 52005	International Space Station Payload Flight Equipment Requirements and Guidelines for Safety-Critical Structures

3.0 MDCA OVERVIEW

3.1 Payload Description

The Multi-user Droplet Combustion Apparatus is a mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of droplet combustion in a microgravity environment. MDCA will be integrated into the Fluids and Combustion Facility's (FCF) Combustion Integrated Rack (CIR) onboard the International Space Station (ISS). Currently, the Flame Extinguishment Experiment (FLEX) and the Flame Extinguishment Experiment 2 (FLEX-2) have been selected for use in the MDCA.

3.2 MDCA Mission

The MDCA mission is to perform microgravity experiments to study fire suppression/flame extinguishment investigations using the CIR. The MDCA will allow investigation using pure liquid fuels and mixtures of liquid fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for measurement and observation of flame extinguishment processes that cannot be made in sustained gravity. The modular design of the MDCA allows for flexibility in configuring for specific experiments and easy maintainability of the hardware. This hardware will provide the apparatus for dispensing, deploying and igniting fuel droplets. When new droplet experiments are proposed, the additional experiment specific hardware will be provided. This requirements document reflects such a case, including only the requirements of the FLEX-2 investigation.

3.3 FLEX-2 Requirements Flow

Figure 1 represents the requirements flow for all of the FLEX-2 requirements. The requirements in the Science Requirements Document (SRD) are agreed upon by the FLEX-2 Principle Investigators (PI's), and this document serves as the top level source of those requirements necessary to achieve the FLEX-2 science objectives. Derived requirements come from engineering requirements and are set by the FLEX-2 Project Team. These two sets of requirements are incorporated into this the FLEX-2 Engineering Requirements Document (FLX-REQ-0001).

Safety requirements are contained in the updated FLEX-2 safety review hazard reports from the Payload Safety Review Panel (PSRP). The carrier requirements are found in the CIR/FLEX-2 Interface Control Document (FCF-ICD-CIR-MDCA).

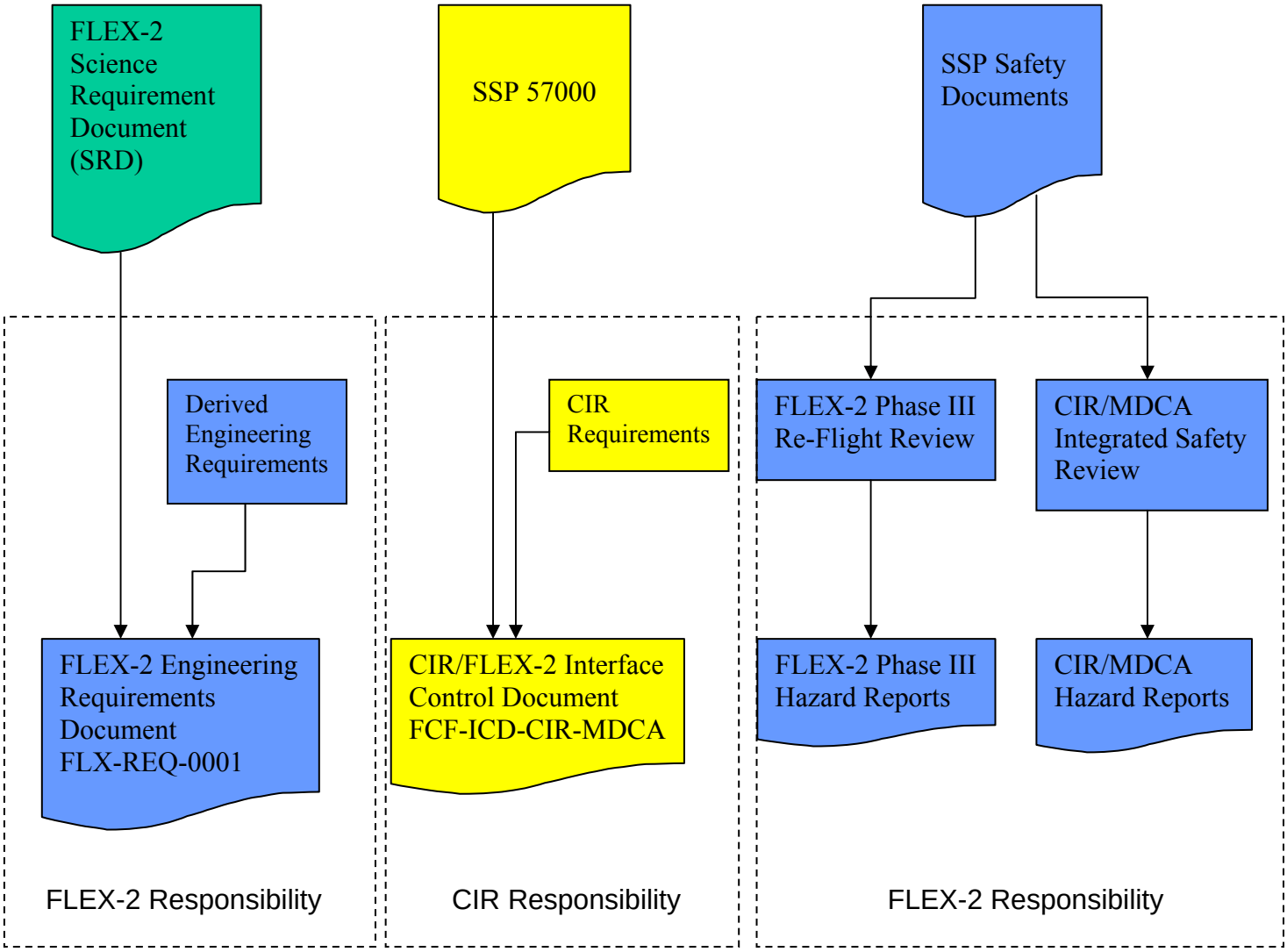


Figure 1 FLEX-2 Requirements Flow

3.4 MDCA Hardware Description

The FLEX-2 investigation will be using the existing MDCA hardware and software. MDCA should be considered the platform on which the FLEX-2 experiments are carried out. The MDCA hardware consists of three hardware domains: the Chamber Insert Assembly (CIA), the Avionics Package, and MDCA Diagnostics. The CIA provides the mechanism for the combustion event. The Avionics Package is the center of the control and power distribution for the FLEX-2. The FLEX-2 hardware is discussed in detail below. If more detail is desired, please refer to MDC-DOC-1364, MDCA Baseline Systems Description.

3.4.1 MDCA Chamber Insert Assembly

The MDCA Chamber Insert Assembly provides all the necessary hardware and interfaces to perform science on one platform. Most experimental functions occur within the CIA structure and many associated systems are found on the CIA. The CIA consists of the Internal Apparatus (IA), which is the functional hardware and the Experiment Mounting Structure (EMS) upon which the IA is structurally based. Other subsystems found on the CIA include the Droplet Dispensing System, the Droplet Deployment System, the Fiber Support System, and the Ignition System.

3.4.1.1 Experiment Mounting Structure

The EMS provides the main structure for the CIA and acts as the primary mechanical interface with the CIR combustion chamber. It consists of two endplates, the support structure, and the removable shrouds to protect the IA during crew activity.

3.4.1.2 Internal Apparatus

The IA provides the primary mounting platform for all experiment specific hardware and functions such as droplet dispensing, deployment and ignition. The IA encompasses all motors used in MDCA, as well as a cooling plate, radiometers, and some avionics hardware. The IA is also available for future add-on hardware for different PIs.

3.4.1.2.1 Fuel Dispenser System

There are two separate fuel dispenser systems on the CIA. Each fuel dispenser system consists of a motor housing and a replaceable reservoir. The motor housing is permanently fixed to the IA and contains the motor, encoder, and lead screw assembly. The encoder verifies position of the motor and allows ground operators to keep account of fuel usage. The replaceable reservoir is composed of the syringe, the latching solenoid valve, the reservoir housing, and a transparent Lexan cover. The replaceable reservoir assembly is located on top of the motor housing. The syringe is covered in a Tygon sleeve and contained in aluminum housing. The syringe is connected directly to a fuel isolation, latching, solenoid valve. The polytetraflouroethylene (PTFE) fuel line tubing is attached to the valve with a removable fitting to disconnect the

deployment needles from the fuel system. Fuel is dispensed by moving the lead screw using a stepper motor, which in turn pushes the plunger on the syringe.

3.4.1.2.2 Deployment System

The deployment system consists of two deployment needles made from .010-inch O.D. stainless steel tubing supported by a ceramic sleeve. Each needle tip is flared to aid in the droplet deployment. The needles are mounted to individual rotary servomotors with a specially designed coupling that allows replacement of the needles and maintains the precise alignment of the system. The motors rotate the needles to a pre-determined location where they are nearly tip-to-tip. Metered fuel from the dispenser is routed via the PTFE tubing to one needle forming a droplet of very precise volume. The deployment motors then actuate again, creating a slight gap between the needles. This causes the droplet to center itself between the needle tips and minimizes the contact area with the needles. Finally, the needles are rapidly accelerated apart, deploying the droplet off the needle tips. The rapid acceleration leaves the droplet in place with little or no residual motion.

3.4.1.2.3 Fiber Support System

The fiber mechanism consists of a 235 mm long, 79μm diameter silicon carbide fiber fastened between two arms with epoxy to a tension of 15000 psi. There is a succession of small stainless steel beads secured to the fiber at 12mm intervals. The use of the fiber mechanism allows fuel droplets to be deployed onto, and thus tethered to, the beaded fiber at specific locations during experimental burn and observation. The beads act to constrain the position of the droplets on the fiber, which is especially necessary for the FLEX-2 droplet linear array experiments, where two droplets are burned simultaneously. When used, the fiber is rotated 68° into the deployment area using a rotary stepping gear motor. To deploy a linear array of droplets and to provide a clean portion of the fiber for successive burns, linear translation of the fiber through the deployment zone is achievable using a similar gear motor driving a pinion gear attached to a rack on the fiber mechanism assembly. This translation can also be used to provide a low velocity flow over a single burning droplet.

3.4.1.2.4 Ignition System

The ignition system uses the original MDCA configuration for single droplet ignition and for the array experiments, modified dual droplet igniter tips are employed. The original MDCA ignition system consists of two hot wire igniters opposed 180°, controlled by individual actuators. These ignition wires are 4mm diameter loops of Kanthal A-1 wire, 30 AWG. These loops are mounted into ceramic tubes and comprise the igniter tip assembly. This assembly has two pins that mate with sockets on the igniter-mounting interface. This interface is fixed to the igniter motors, which move the igniters into the deployment area linearly in precise steps and quickly retract the igniters after ignition. The MDCA igniter tips are designed to be crew replaceable and for the array experiments these will be replaced with the dual droplet igniter tips, which are based on the original MDCA design and comply with all MDCA requirements.

3.4.1.2.5 Cooling Plate System

A cooling plate is provided to conductively cool the IA. The tubing is a stainless steel 3/8-inch O.D. with .028-inch wall thickness. CIR compatible quick disconnect coupling fittings will connect the cold plate to the EMS endplate. A flexible conductive lined Teflon hose with stainless steel braiding will provide the coupling from the MDCA CIA to the CIR Interface Resource Ring (IRR). Mating Preece connectors will be used for this interface with CIR's IRR. CIR's color coding scheme will be maintained by MDCA for the inlet and outlet connectors.

The cooling system will circulate ISS cooling water flow controlled by CIR to satisfy FLEX-2 requirements. Prior to launch, the cooling system will be pre-filled to capacity with cooling water. A gaseous nitrogen charged accumulator bottle installed inline will provide automatic CIA protection from fluid over expansion in the system specifically during storage periods where ambient temperatures may significantly change.

3.4.1.2.6 Radiometers

The Radiometers will provide relative measure of flame radiation and emissions through the combustion event. The Radiometer Package consists of two Radiometers contained in a single housing, mounted on the CIA, providing a direct view of the combustion event. The Broad-Band Radiometer will measure total flame energy from combustion over a wavelength range covering 0.6 μm – 5.0 μm, while the Narrow-Band Radiometer will provide spectrally filtered measurements for specific H_20 emissions at 1.87 μm, using a narrow bandpass filter. Power to the Radiometer Package is provided by MDCA Avionics.

3.4.2 MDCA Avionics

The Avionics Package provides power conditioning and power distribution, system control, data acquisition, and operator interface for the MDCA. The Avionics Package will house the main electronics for the MDCA hardware and will be mounted on the CIR Optics Bench.

3.4.2.1 Data and Control

A processor is responsible for automating the sequencing of operational events along with acquiring, downlinking, and storing experiment data. Experiment control and sequencing will be automated by the MDCA software, which is resident on the main processor. The main processor communicates with the CIR Input/Output Processor (IOP). The interface to the CIR IOP is necessary for transferring commands, downlinked data, and images and to receive the ISS video and health and science (H&S) data.

3.4.2.2 Power Control and Distribution

MDCA receives its direct current, 28-Volt power from the CIR's Electrical Power Control Unit (EPCU) through the Avionics Package. In the Avionics Package, the power is conditioned by filters and dc-dc converters for use by the processor, eight Axis Motor Controller, A/D and Signal Conditioning Boards, and CIA interface board.

3.4.3 MDCA Diagnostics

MDCA Diagnostics provide acquisition and storage of all optical information concerning droplet deployment and combustion.

3.4.3.1 Color Camera Package

The MDCA Color Camera Package will provide an overhead view of the combustion event. The Color Camera Package consists of a Sony FCB-EX470L CCD camera, which includes a built-in Zoom Lens with RS-232 controllable zoom and focus functions. The camera is completely enclosed in a housing container and mounted to the CIA upper structure, internal to the chamber. An aluminum broadband coated turning mirror is used to fold the view from the experiment to the color camera. The Color Camera Package analog video output is converted to twisted pair, via the Color Camera Interface Package, which then routes the output to the CIR Image Processing and Storage Unit (IPSU-A) for viewing and storage. Power to the Color Camera Package is supplied by MDCA Avionics.

3.4.3.2 White Light Illumination

The White Light Illumination Package supports the Color Camera Package prior to and following the experiment event, to provide direct overhead lighting of the droplet deployment area during dispensing, deployment and ignition for troubleshooting purposes. The White Light Illumination is powered off upon droplet ignition, as the experiment provides its own light. The package consists of two clusters of nine White Light LEDs mounted on the CIA upper structure. Power to the White Light Illumination is supplied by MDCA Avionics.

3.5 Interfaces

3.5.1 External Interfaces

3.5.1.1 Launch Configuration Interfaces

FLEX-2 provided hardware is non-operational for launch. FLEX-2 hardware will be stowed in standard containers provided by FLEX-2 and will only have physical and structural interfaces with the containers.

3.5.1.2 On-Orbit External Interfaces

FLEX-2 has no direct external interfaces on-orbit. All interfaces with the FCF, US LAB, ISS and the ground are indirectly through CIR.

3.5.2 Internal Interfaces

The internal interfaces between MDCA and CIR are listed below and the requirements for these interfaces will be defined in FCF-ICD-CIR-MDCA, Multi-User Droplet Combustion Apparatus Interface Control Document.

- Structural/Mechanical and Stowage
- Electrical Power
- Command and Data Handling
- Payload and Diagnostics
- Thermal Control
- Exhaust Vent
- FOMA Gas Delivery
- Environmental

4.0 REQUIREMENTS

This section details all science and derived engineering requirements that have been levied upon FLEX-2. Verifications for these requirements are documented in the FLEX-2 SEVP (FLX-PLN-0002).

4.1 Science Requirements

This section is a flowdown of the science requirements from the FLEX-2 SRD dated December 6, 2009. Upon being signed, this FLEX-2 ERD becomes the formal FLEX-2 project requirement document.

4.1.1 Experimental Fuels - Alcohol

FLEX-2 shall require the use of ethanol.

4.1.2 Experimental Fuels – Hydrocarbon

FLEX-2 shall require the use of decane.

4.1.3 Experimental Fuels – Alcohol Mixture

FLEX-2 shall require the use of a propanol/glycerol mixture.

4.1.4 Experimental Fuels – Hydrocarbon Mixture 1

FLEX-2 shall require the use of a heptane/hexadecane mixture.

4.1.5 Experimental Fuels – Hydrocarbon Mixture 2

FLEX-2 shall require the use of a heptane/octane mixture.

4.1.6 Experimental Fuels – Other Mixture

FLEX-2 shall require the use of a decane/propylbenzene mixture.

4.1.7 Fuel Purity

FLEX-2 shall use fuels of research grade with purity greater than 99.5% by volume.

4.1.8 Fuel Quality

FLEX-2 fuels shall be anhydrous prior to loading into fuel reservoirs.

4.1.9 Fuel Degassing

FLEX-2 pure fuels shall be degassed prior to loading into fuel reservoirs.

4.1.10 Fuel Quality Assurance

Certification of test samples shall be provided prior to flight.

4.1.11 Fuel Temperature

The fuel temperature at the start of each test shall be in the range of 18 ºC – 27 ºC.

4.1.12 Droplet Size Tolerance

FLEX-2 shall dispense droplets with diameters of 1.5 – 5 mm with a tolerance ± 0.25 mm.

4.1.13 Droplet Size Reproducibility

FLEX-2 shall dispense droplets with diameters of 1.5 – 5 mm with a reproducibility of ± 5%.

4.1.14 Fiber Size

FLEX-2 shall provide a support fiber with a diameter < 100 μm.

4.1.15 Fiber Material Thermal Diffusivity

FLEX-2 shall provide a fiber material with a thermal diffusivity no greater than 2.3 cm^2/s.

4.1.16 Fiber Material Emissivity

FLEX-2 shall provide a fiber material with an emissivity comparable to that of 80 micron. diameter Silicon Carbide.

4.1.17 Droplet Anchoring Capability

FLEX-2 shall provide a support fiber capable of anchoring a fuel droplet at a fixed location by use of bead structures.

4.1.18 Droplet Arrays

FLEX-2 shall provide the capability to create droplet arrays in a binary linear configuration with bead spacing of 12mm and 24mm.

4.1.19 Bead Material

FLEX-2 shall ensure that the beads are made of an impervious material.

4.1.20 Bead Shape

FLEX-2 shall ensure that the beads are of a shape that is as close to spherical as possible.

4.1.21 Bead Size

FLEX-2 shall ensure that the beads have a diameter that is as small as possible, but of sufficient size as to allow consistent droplet deployment and positioning for the full range of initial droplet diameters.

4.1.22 Bead Surface

FLEX-2 shall ensure that the surface of the beads and the fiber interface contain no pores or areas where gas can be trapped.

4.1.23 Bead Spacing

FLEX-2 shall provide the capability to create droplet arrays in a binary linear configuration with a bead spacing of 12mm.

4.1.24 Ignite in Flow

FLEX-2 shall provide the capability to ignite a droplet in a quiescent atmosphere and with gas flowing past.

4.1.25 Igniter Material

FLEX-2 shall provide hot wire igniters as the droplet ignition source.

4.1.26 Igniter on Time

FLEX-2 shall provide the capability to power on the igniters for between 1 to 5 second durations.

4.1.27 Ignition Timing

FLEX-2 shall wait 10 seconds after deployment of a droplet before initiating the ignition sequence to allow for droplet deployment induced oscillations to damp adequately.

4.1.28 Ignition Disturbance

FLEX-2 shall provide the command controllable ability to apply the minimum necessary energy input to the droplet during ignition.

4.1.29 Igniter Position Control

FLEX-2 shall provide capability to reposition the igniter tip between tests to 1 – 5 initial droplet diameters from the droplet surface, ± 0.5 mm.

4.1.30 Igniter Retraction

FLEX-2 shall provide the capability to retract the igniter tips from the field of view of the droplet view diagnostic after ignition of the droplet.

4.1.31 Igniter Control

FLEX-2 shall provide the capability to change the igniter position, power, and duration of operation by command.

4.1.32 Hardware Exclusion

FLEX-2 shall provide for removal of all objects (except the fiber) inside the spherical boundary of a radius of 10 initial droplet diameters from the deployment site after ignition during all tests.

4.1.33 Non-Quiescent Test Point Timing

When running a non-quiescent test point, the translation of the support fiber shall begin at the same time as igniter retraction.

4.1.34 Chamber Atmosphere Pressure

FLEX-2 shall provide ambient pressure ranges from 0.5 – 3 atm, with a tolerance of ± 0.05 atm.

4.1.35 Chamber Atmosphere Pressure Rise

FLEX-2 shall maintain the chamber pressure throughout any given test point to within ± 10% of the initial chamber pressure.

4.1.36 Oxygen Mole Fractions

FLEX-2 shall provide oxygen mole fractions ranging from 0.1 to 0.4 with a tolerance on the initial charge of ± 0.005 for O_2.

4.1.37 Chamber Atmosphere Temperature

FLEX-2 shall provide ambient temperature within the range of 18 – 27 °C.

4.1.38 Atmospheric Mixing

FLEX-2 shall provide well-mixed chamber gasses before ignition.

4.1.39 Atmospheric Quiescence

FLEX-2 shall provide quiescent gasses prior to ignition.

4.1.40 Atmospheric Humidity

Atmosphere shall have a Relative Humidity of < 10% for all tests.

4.1.41 Chamber Atmosphere Test Set 1

FLEX-2 shall provide gas mixtures of O_2/N_2.

4.1.42 Chamber Atmosphere Test Set 2

FLEX-2 shall provide gas mixtures of O_2/He.

4.1.43 Flow Field Production

FLEX-2 shall provide relative motion between the drop and the ambient gas by translating the fiber.

4.1.44 Flow Field Velocity Ranges

FLEX-2 shall provide flow velocity ranges from 0.1 to 5cm/s with an accuracy of ±1%.

4.1.45 Flow Field Control

FLEX-2 shall provide for control of the acceleration and deceleration of the flow field.

4.1.46 Test Point Initiation

FLEX-2 shall wait at least two (2) minutes after a chamber atmospheric fill before beginning a test point to ensure the gas, temperature and pressure to stabilize.

4.1.47 Display of Diagnostic Views

FLEX-2 shall provide the capability to downlink the color camera view in near real time.

4.1.48 Downlink of Chamber Temperature Data

FLEX-2 shall provide near real time downlink of the chamber temperature data.

4.1.49 Downlink of Chamber Pressure Data

FLEX-2 shall provide near real time downlink of the chamber pressure data.

4.1.50 Atmospheric Fuel Vapor

The test atmosphere prior to each test point shall have fuel vapor mole fraction of < 0.005.

4.1.51 Atmospheric Purity

The test atmosphere prior to each test shall consist of < 0.02 mole fraction (for each product species) CO, CO_2, and any other by-products of combustion.

4.1.52 G-Level

FLEX-2 shall ensure acceleration levels are < 1 x 10^{-4} m/s^2 on three (3) axes.

4.1.53 Microgravity Measurement

FLEX-2 shall provide capability to measure acceleration near the CIR with an accuracy of 1 x 10^{-6} m/s^2. Frequency range shall be 0.01 – 125 Hz. Sampling rate shall be 1 – 5 times the frequency.

4.1.54 Diagnostics Imaging

All MDCA diagnostics shall acquire images pre-ignition, during ignition, at droplet extinction and post-extinction.

4.1.55 Droplet Imaging Orientation

FLEX-2 shall provide a droplet view with the focal plane parallel to the support fiber and coplanar with the droplet.

4.1.56 Droplet Imaging FOV

FLEX-2 shall provide FOV of 3.0 cm diameter centered about the deployment site for the droplet view.

4.1.57 Droplet Imaging Spatial Resolution

FLEX-2 shall provide spatial resolution of 30 μm over the entire FOV for the droplet view.

4.1.58 Droplet Imaging Temporal Resolution

FLEX-2 shall provide a nominal minimum framing rate of 30 fps for the droplet view.

4.1.59 Droplet Imaging Temporal Resolution Additional Capability

The FLEX-2 droplet view camera shall have the ability to use a framing rate of between 60 to 100 fps at a resolution of 60 μm.

4.1.60 Droplet Imaging DOF

FLEX-2 shall provide a DOF ≥ 3.0 cm for the droplet view.

4.1.61 Droplet Imaging Illumination

FLEX-2 shall provide capability to back light the droplet with a uniform, collimated light source of sufficient intensity to allow the droplet's edges to be clearly detected.

4.1.62 OH Flame Imaging Orientation

FLEX-2 shall provide a flame view with a focal plane parallel to the support fiber.

4.1.63 OH Flame Imaging FOV

FLEX-2 shall provide FOV ≥ 5.0 cm centered about the deployment site for the OH flame view.

4.1.64 OH Flame Imaging Emission

FLEX-2 shall provide color imaging detection with emission centered for wavelengths of 310 nm (± 5 nm).

4.1.65 OH Flame Imaging Spatial Resolution

FLEX-2 shall provide a spatial resolution of 150μm over the entire FOV for the OH flame view.

4.1.66 OH Flame Imaging Temporal Resolution

FLEX-2 shall provide framing rate of 30 fps for the OH flame view.

4.1.67 OH Flame Imaging DOF

FLEX-2 shall provide a DOF no less than 5mm for the OH flame view.

4.1.68 OH Flame Imaging Gain

The gain on the OH flame imager shall be adjustable prior to each test point.

4.1.69 Color Imaging Orientation

FLEX-2 shall ensure the MDCA color camera provides a full view of the droplet and flame with the focal plane parallel to the fiber providing viewing of the droplet dispensing and deployment sequence.

4.1.70 Color Imaging FOV

FLEX-2 shall provide a FOV $\geq$ 5.0 cm diameter, positioned so the entire flame is imaged for the MDCA color camera.

4.1.71 Color Imaging Spatial Resolution

FLEX-2 shall provide a spatial resolution of 100μm for the MDCA color camera.

4.1.72 Color Imaging Temporal Resolution

FLEX-2 shall provide framing rate of 30 fps for the MDCA color camera.

4.1.73 Color Imaging DOF

FLEX-2 shall provide a DOF of no less than 5.0 mm for the MDCA color camera.

4.1.74 Color Imaging Zoom

FLEX-2 shall provide a zoom capability with the MDCA color camera, upon request of the PI, in order to adjust the FOV.

4.1.75 Narrow Band Radiative Emission

FLEX-2 shall provide a narrow band radiometer to measure radiant emissions in the luminous and non-luminous flames at a wavelength in the H2O band centered at 1.87 μm.

4.1.76 Narrow and Broad Band Radiative Emission Sampling Rate

FLEX-2 shall provide a narrowband and broadband radiometer sampling rate of 30 Hz.

4.1.77 Narrow Band and Broad Band Radiative Emission Placement

FLEX-2 shall place the narrowband and broadband radiometer at least 15 cm from the droplet.

4.1.78 Narrow Band and Broad Band Radiometer Isolation

FLEX-2 shall shield the narrowband and broadband radiometers from igniter wire radiation, both reflected and direct.

4.1.79 Narrow Band and Broad Band Radiative Emission Accuracy

The FLEX-2 narrow band and broad band radiative emission accuracy shall be 5% of full scale incident radiation.

4.1.80 Narrow Band and Broad Band Radiative Emission Response Time

The FLEX-2 narrow band and broad band radiative emission shall have a response time constant of less than 40 ms.

4.1.81 Narrow Band and Broad Band Radiative Emission FOV

FLEX-2 shall provide a narrowband and broadband radiometer FOV of 65 mm in diameter, centered about the deployment site.

4.1.82 Broad Band Radiative Emission

FLEX-2 shall employ a broad band radiometer to measure radiant emissions in the luminous and non-luminous flames in the wavelength range of 0.6-5.0 μm.

4.1.83 Soot Volume Fraction Imaging Orientation

FLEX-2 shall provide a full view of the droplet and flame with a preferred orientation such that the focal plane is parallel to the fiber and coplanar with the droplet.

4.1.84 Soot Volume Fraction Backlight

FLEX-2 shall provide a collimated backlit soot volume fraction diagnostic with the backlight at a wavelength of 660 nm and a Full Width Half Maximum of 5 nm.

4.1.85 Soot Volume Fraction Temporal Resolution

FLEX-2 shall provide a soot volume fraction diagnostic with a framing rate of ≥ 15 fps.

4.1.86 Soot Volume Fraction Spatial Resolution

FLEX-2 shall provide a soot volume fraction diagnostic with a resolution of 50 μm over the entire FOV.

4.1.87 Soot Volume Fraction Field of View

FLEX-2 shall provide a soot volume fraction diagnostic with a full view of the droplet and flame, having a FOV ≥ 5.0 cm centered on the droplet deployment site.

4.1.88 Soot/Fiber Temperature Imaging Orientation

FLEX-2 shall provide a full view of the droplet and flame with a preferred orientation such that the focal plane is parallel to the fiber and coplanar with and centered on the droplet.

4.1.89 Soot/Fiber Temperature Field of View

FLEX-2 shall provide a soot /fiber temperature diagnostic with a full view of the droplet and flame, having a FOV ≥ 5.0 cm centered on the droplet deployment site.

4.1.90 Soot/Fiber Temperature Temporal Resolution of Liquid Crystal Tunable Filter

FLEX-2 shall provide the capability of alternating the two tuned wavelengths of the filter, 700 nm and 800 nm, at a frequency that results in even numbered frames being exposed to one of the two wavelengths, and odd frames exposed to the other. This shall result in an effective frame rate of at least 7fps for each wavelength.

4.1.91 Soot/Fiber Temperature Spatial Resolution

FLEX-2 shall provide a soot /fiber temperature diagnostic with a resolution of 50 μm over the entire FOV.

4.1.92 Flow Velocity Measurement

FLEX-2 shall record all flow velocity measurements.

4.1.93 Flow Field Sampling Rate

FLEX-2 shall provide for the sampling of the translated motion of the FLEX-2 support fiber at a rate ≥ 30 Hz.

4.1.94 Chamber Temperature Measurement Sampling Frequency

FLEX-2 shall provide a sampling frequency of the chamber temperature of at least 10 Hz.

4.1.95 Chamber Pressure Measurement Sampling Frequency

FLEX-2 shall provide a sampling frequency of the chamber pressure of at least 10 Hz.

4.1.96 Pressure Measurement Accuracy

FLEX-2 shall provide pressure measurement accuracy to within ± 0.01 atm in the range of 0.5-3.0 atm.

4.1.97 Temperature Measurement Accuracy

FLEX-2 shall provide ambient temperature measurement accuracy to within ±0.5 C in the range of 18-27º C.

4.1.98 Synchronization of Measurements

FLEX-2 shall reference time measurements to Greenwich Mean Time (GMT) and have all data time stamped and synchronized to within ± 0.03 sec.

4.2 Derived Requirements

This section specifies the additional derived requirements needed to support the training, testing, operations, design, and development of the FLEX-2 project within the MDCA and CIR carriers.

4.2.1 Fuel Dispenser System

The following requirements refer specifically to the MDCA/FLEX-2 fuel dispenser system.

4.2.1.1 Fuel Mixture Ratio Accuracy

FLEX-2 shall develop a procedure to ensure that accurate fuel mixtures are loaded into the fuel reservoirs. Mixture ratios shall be accurate to within ±0.5%.

4.2.1.2 Fuel Material Compatibility

The materials exposed to the fuels within the dispensing system shall be compatible with FLEX-2 identified fuels as follows:

1. Ethanol
2. Decane
3. Heptane
4. Hexadecane
5. Octane
6. Propylbenzene
7. Glycerol
8. Propanol

4.2.1.3 Purity Degradation

Purity degradation is in direct correlation to the fuels' exposure to outside environments. Fuel purity degradation shall be ensured by sealing the fuel reservoir assembly to a maximum leak rate of 1×10^{-6} scc/sec of gaseous helium.

4.2.2 Atmospheric Conditions

The following requirements refer specifically to FLEX-2 chamber atmospheric conditions.

4.2.2.1 CO Removal

FLEX-2 shall provide the means to remove CO from the chamber atmosphere to the acceptable levels detailed in the Science Requirements section, requirement 4.1.51.

4.2.2.2 CO_2 Removal

FLEX-2 shall provide the means to remove CO_2 from the chamber atmosphere to the acceptable levels detailed in the Science Requirements section, requirement 4.1.51.

4.2.2.3 Water Vapor Removal

FLEX-2 shall provide the means to remove water vapor from the chamber atmosphere to the acceptable levels detailed in the Science Requirements section, requirement 4.1.40.

4.2.2.4 Fuel Vapor Removal

FLEX-2 shall provide the means to remove fuel vapor from the chamber atmosphere to the acceptable levels detailed in the Science Requirements section, requirement 4.1.50.

4.2.3 Internal Apparatus

The following requirements refer specifically to the hardware components found on the MDCA Internal Apparatus.

4.2.3.1 Oxygen Compatibility

All components developed for FLEX-2 located in the CIR chamber shall be compatible with 40% oxygen at 3 atm.

4.2.3.2 Igniter Compatibility

FLEX-2 shall provide igniter tips that comply with the existing MDCA igniter drive circuitry. The igniter shall be constructed of 26 AWG Kanthal type A1 wire 75.8 mm long with a total DC resistance of 1 Ω ± 0.250Ω at 20°C.

4.2.3.3 Igniter Hoop Diameter

The igniter hoop diameter shall be able to radiatively heat droplets spaced up to 24mm apart.

4.2.3.4 Fiber Alignment with Needles

The fiber shall be oriented orthogonal to the needle line of action within ± 5 degrees. This value shall be maintained for all replaceable units (when a component is replaced, it shall have the same alignment).

4.2.3.5 Fiber Positioning

The fiber shall be positioned within a spherical segment tolerance band located 0.5 mm above the theoretical center point of deployment, with a radius of 0.25 mm. This value shall be maintained for all replaceable units (when a component is replaced, it shall have the same alignment) and shall not consider bead shape for beaded fibers.

4.2.3.6 Fiber Beads

Fiber beads must not generate localized boiling in the droplet prior to flame contraction. Porous beads are known to generate such disturbances in burning droplets.

4.2.4 Diagnostics

The following requirements refer specifically to the MDCA/FLEX-2 diagnostic packages.

4.2.4.1 Droplet View Filter

FLEX-2 shall provide a HiBMS camera for the backlit droplet view fitted with a narrow band filter compensator with the following properties:

1. A center wavelength of 653 nm and a bandwidth of less than 12 nm.
2. A peak transmission (%T) of at least 80%.
3. Out of Band Blocking of less than 1% over the range of 400nm-1100nm.

4.2.5 Flight Unit Hardware

The following requirements refer specifically to the MDCA Flight Unit Hardware.

4.2.5.1 Flight Unit Packaging, Handling, and Transportation

Packaging, Handling, and Transportation of all flight hardware developed by FLEX-2 shall be in accordance with NPR 6000.1.

5.0 CARRIER CAPABILITIES

Table I lists the capabilities of the CIR/MDCA carriers utilized by the FLEX-2 project in order to achieve its science objectives. These capabilities are documented in the MDCA/FLEX Engineering Requirements Document, MDC-REQ-0005A and have previously been verified by the MDCA/FLEX verification process.

TABLE I. Previously Verified MDCA/FLEX Requirements from MDC-REQ-0005A

Requirement #	Requirement Title
4.3.1.2.1	Determine Change in Oxygen Concentration
4.3.1.2.6	Chamber Atmosphere Mixing
4.3.1.2.7	Chamber Atmosphere Quiescence
4.3.1.3.7	Alignment Verification
4.3.1.3.10	Needle Cover
4.3.1.3.11	Fiber Translation Accuracy
4.3.1.3.13	Fiber Arm Installation/Removal
4.3.1.3.14	Fiber Arm Positioning
4.3.1.3.15	Zone of Exclusion
4.3.1.3.22	Igniter Hoop Alignment
4.3.1.3.28	Needle Symmetry
4.3.1.3.24	Igniter Acceleration
4.3.1.3.26	Needle Positioning Accuracy
4.3.1.3.27	Needle Alignment
4.3.1.3.23	Igniter Positioning Accuracy
4.3.1.4.1	Diagnostics
4.3.1.4.2	MDCA Color Camera Imaging
4.3.1.4.3	MDCA Color Camera Spatial Resolution

4.3.1.4.4	MDCA Color Camera Temporal Resolution
4.3.1.4.5	CIA Deployment Site Alignment
4.3.1.4.6	CIA Rail Bearing Assembly Alignment
4.3.1.4.7	Radiometer Calibration
4.3.1.4.8	Chamber Illumination
4.3.1.4.9	Diagnostics System Availability
4.3.1.5.1	Flight Software
4.3.1.5.2	Software Design Document
4.3.1.5.3	Deployment Motor Synchronicity
4.3.1.5.4	Igniter Motor Synchronicity
4.3.1.5.5	Needle Position for Removal
4.3.1.5.7	Experiment Calculated Data
4.3.1.5.8	Commands and Data
4.3.1.5.9	MDCA Embedded Code
4.3.1.5.11	MDCA Main Computer Time Sync
4.3.1.5.12	Fiber Insertion and Retraction
4.3.1.5.13	Automated Sequence
4.3.1.5.14	Automated Sequence Elements
4.3.1.5.15	Automated Sequence Parameters
4.3.1.5.16	Igniter Position
4.3.1.5.17	Test Point Parameter Limits
4.3.1.5.18	Diagnostic and Illumination Timing
4.3.1.5.23	Health Monitoring Data
4.3.1.5.25	Telemetry Sampling for 1 Hz

4.3.1.5.26	Motor Control
4.3.1.5.27	Software Initialization
4.3.1.5.28	Main Computer Modules
4.3.1.5.29	Hardware Safing and Shutdown
4.3.1.5.38	Needle Motor Drive Parameters
4.3.1.5.31	Dispense Small Increment of Fuel
4.3.1.5.32	MDCA Software Inhibit
4.3.1.5.33	Disable Test Point Inhibit
4.3.1.5.34	Modify the Needle Dispense Positions
4.3.1.5.35	Test Condition Alert
4.3.1.5.36	Message Log
4.3.1.5.37	Droplet Diameter
4.3.1.5.30	Main Computer Shutdown Sequence
4.3.1.5.40	Test Sequence Timing
4.3.1.5.39	Dispenser Motor Driver Parameters
4.3.1.5.41	Main Computer to IOP Transactions
4.3.1.5.42	Main Computer Downlink
4.3.1.7.4	Igniter Power Symmetry
4.3.1.7.6	Avionics System Availability
4.3.1.8	CIA Assembly
4.3.1.8.2	CIA System Availability
4.3.1.10	CIA Volume
4.3.1.11	Flight Unit Hardware Availability

APPENDIX A ACRONYMS AND ABBREVIATIONS

A.1 Scope

This appendix lists the acronyms and abbreviations used in this document.

A.2 List of Acronyms and Abbreviations

ARIS.....Active Rack Isolation System
AS .....Automated Sequence
AWG.....American Wire Gauge
CIA .....Chamber Insert Assembly
CIL.....Critical Items List
CIR .....Combustion Integration Rack
CM.....Configuration Management
CPP.....Contractor Project Plan
dc .....Direct Current
DCM .....Diagnostic Control Module
DOF.....Depth of Field
EMI.....Electromagnetic Interference
EMS .....Experiment Mounting Structure
EPCU .....Electrical Power Control Unit
ERD.....Engineering Requirements Document
FCF .....Fluids and Combustion Facility
FLEX .....Flame Extinguishment Experiment
FLEX-2 .....Flame Extinguishment Experiment 2
FMEA .....Failure Modes and Effects Analysis
FOMA.....Fuel/Oxidizer Management Assembly
FOV.....Field of View
FPS .....Frames per Second
FWHM .....Full Width-Half Maximum
GMT .....Greenwich Mean Time
GRC .....Glenn Research Center
GSE.....Ground Support Equipment
H&S.....Health and Science
IA.....Internal Apparatus
ICD .....Interface Control Document
IFOV.....Instantaneous Field of View
IOP .....Input/Output Processor
IRR .....Interface Resource Ring
ISPR.....International Standard Payload Rack
ISS .....International Space Station
IVA .....Intra-Vehicular Activity
LED .....Light Emitting Diode
LOS .....Loss of Signal
MAPTIS.....Materials and Processes Technical Information Systems

MDCA.....Multi-user Droplet Combustion Apparatus
MDP .....Maximum Design Pressure
MPLM.....Multi-Purpose Logistics Module
MRDOC.....Microgravity Research Development and Operations Contract
NASA .....National Aeronautics and Space Administration
NSTS.....National Space Transportation System
O.D.....Outer Diameter
PFE .....Portable Fire Extinguisher
PI.....Principal Investigator
PL.....Payload
PTFE .....Polytetraflouroethylene
QD.....Quick Disconnect
RH .....Relative Humidity
SCC.....Stress Corrosion Cracking
SEE .....Single Event Effect
SMAC.....Spacecraft Maximum Allowable Concentrations
SpaceDOC .....Space Flight Systems Development & Operations Contract
SRD.....Science Requirements Document
SSC.....Station Support Computer
TBD .....To Be Determined
TSC .....Telescience Support Center
UML.....Universal Mounting Location
USL .....United States Laboratory
UV .....Ultra-Violet
WBS……Work Breakdown Structure

APPENDIX B DEFINITIONS

B.1 Scope

This appendix defines special terminology used in this document.

B.2 List of Definitions

Chamber Insert Assembly (CIA) – Configured equipment placed inside the combustion chamber to perform an experiment. Usually consists of an experiment mounting structure and an internal apparatus.

Critical Items List (CIL) – For critical failure modes, the CIL contains specific rationale, which justifies the retention of the critical item in the system.

Experiment Mounting Structure (EMS) – Hardware that provides the structural interface from the combustion chamber to the internal apparatus.

Failure Modes and Effects Analysis (FMEA) – Procedure by which each credible failure mode is analyzed (failure mode occurrence, methods of detection the failure, corrective action) to determine the effects on the system and to classify each potential failure mode in accordance with the severity of its effect.

Internal Apparatus (IA) – The IA provides the primary mounting platform for all experiment specific hardware and functions such as droplet dispensing, deployment and ignition. The IA encompasses all motors used in MDCA, as well as a cooling plate, radiometers and some avionics hardware. The IA is also available for future add-on hardware for different PIs.

Telescience Support Center (TSC) – The NASA ground facility that provides the capability to execute ground support operations of on-orbit ISS and Space Shuttle Payloads from the Glenn Reseach Center (GRC).